Computer Organization and Architecture (CSL251) 2025-26 W
LAB-14
Date: 20.04.2026

1. Verilog code to design a block that "selects 1 among 2 n input lines and connects it the 1-bit output line, based on the value of n-Bit input control lines". For example, if value of n=3 and then value of control line is 000 (or 111) the 1 st (8 th) input line is connected to the output. For implementation, consider 3 select lines (i.e., n=3), where based on the value of n, the corresponding input line is connected to the output.

Implement using the following options

If-else

Design:

```
module mux_example(
  input [2:0] sel,
  input i0, i1, i2, i3, i4, i5, i6, i7,
  output reg y
);
  always @(*) begin
    if (sel == 3'b000)
      y = i0;
    else if (sel == 3'b001)
      y = i1;
    else if (sel == 3'b010)
      y = i2;
    else if (sel == 3'b011)
      y = i3;
    else if (sel == 3'b100)
      y = i4;
    else if (sel == 3'b101)
      y = i5;
    else if (sel == 3'b110)
      y = i6;
    else if (sel == 3'b111)
      y = i7;
    else begin
      y = 1'bx; // undefined if invalid
      $display("Invalid sel input: %b", sel);
    end
  end
endmodule
```

Testbench:

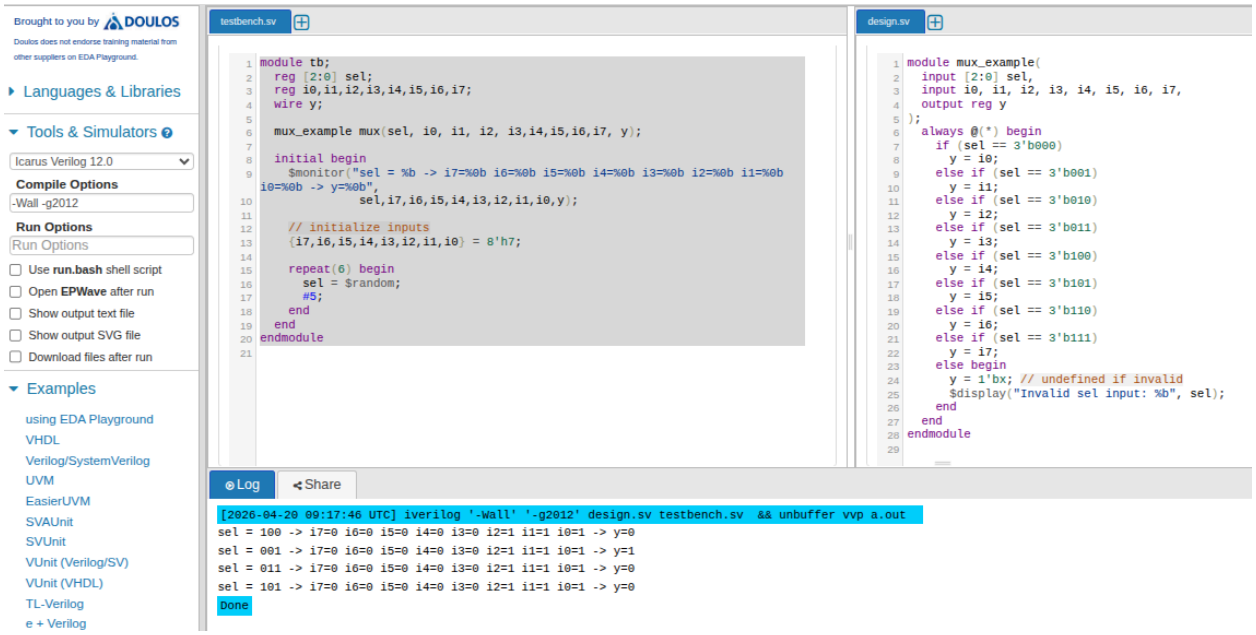
```
module tb;
  reg [2:0] sel;
  reg i0,i1,i2,i3,i4,i5,i6,i7;
  wire y;

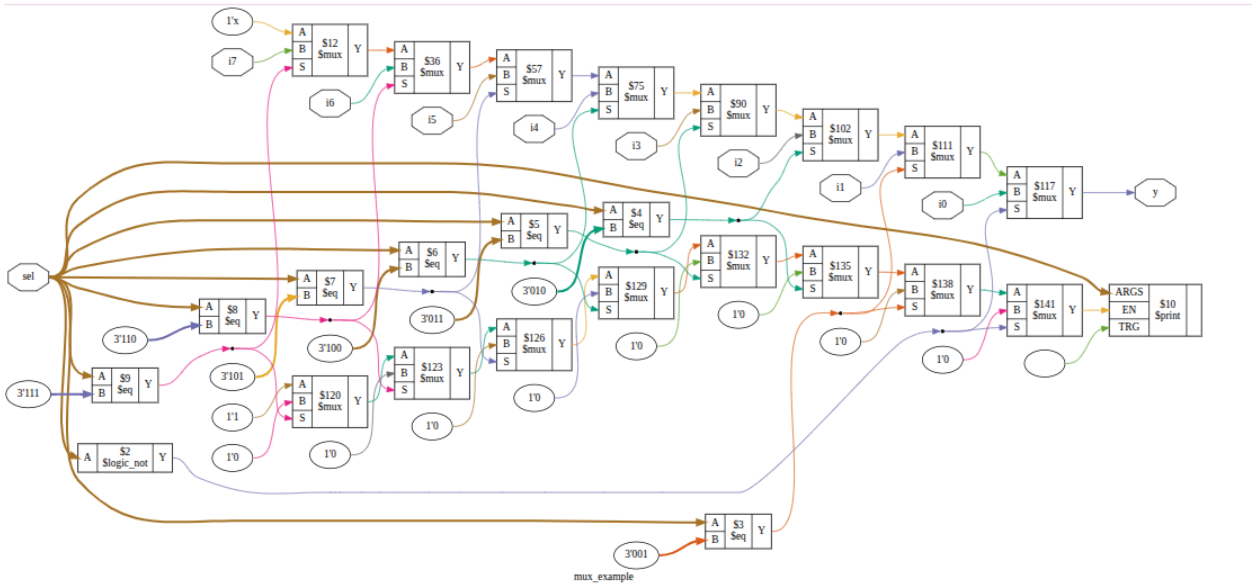
  mux_example mux(sel, i0, i1, i2, i3,i4,i5,i6,i7, y);

  initial begin
    $monitor("sel = %b -> i7=%0b i6=%0b i5=%0b i4=%0b i3=%0b i2=%0b i1=%0b i0=%0b ->
y=%0b",
             sel,i7,i6,i5,i4,i3,i2,i1,i0,y);

    // initialize inputs
    {i7,i6,i5,i4,i3,i2,i1,i0} = 8'h7;

    repeat(6) begin
      sel = $random;
      #5;
    end
  end
endmodule
```





2. Design a Verilog module to generate an **odd parity bit** for a 4-bit binary input.

- The module takes a **4-bit input** `number`
- It produces a **1-bit output** `parity`
- The output `parity` should be such that the **total number of 1's (including the parity bit) is odd**

Example:

- If `number = 4'b0000` → number of 1's = 0 (even) → `parity = 1`
- If `number = 4'b0001` → number of 1's = 1 (odd) → `parity = 0`
- If `number = 4'b1111` → number of 1's = 4 (even) → `parity = 1`

Design:

```
module odd_parity(
    input [3:0] number,
    output reg parity
);
```

```
reg [2:0] parity_sum;

always @(*) begin
    parity_sum = number[0] + number[1] + number[2] + number[3];

    if (parity_sum[0] == 0)
        parity = 1;
    else
        parity = 0;
end

endmodule
```

Testbench:

```
`timescale 1ns/1ps

module odd_parity_tb;

    reg [3:0] number;
    wire parity;

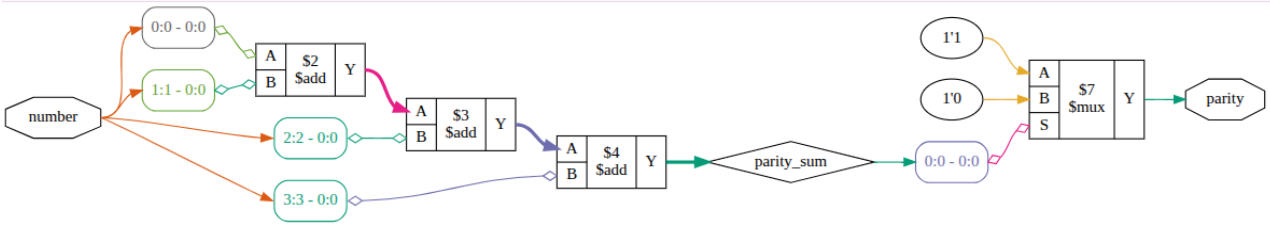
    odd_parity dut (
        .number(number),
        .parity(parity)
    );

    initial begin
        $monitor("Time=%0t | number=%b | parity=%b",
                $time, number, parity);

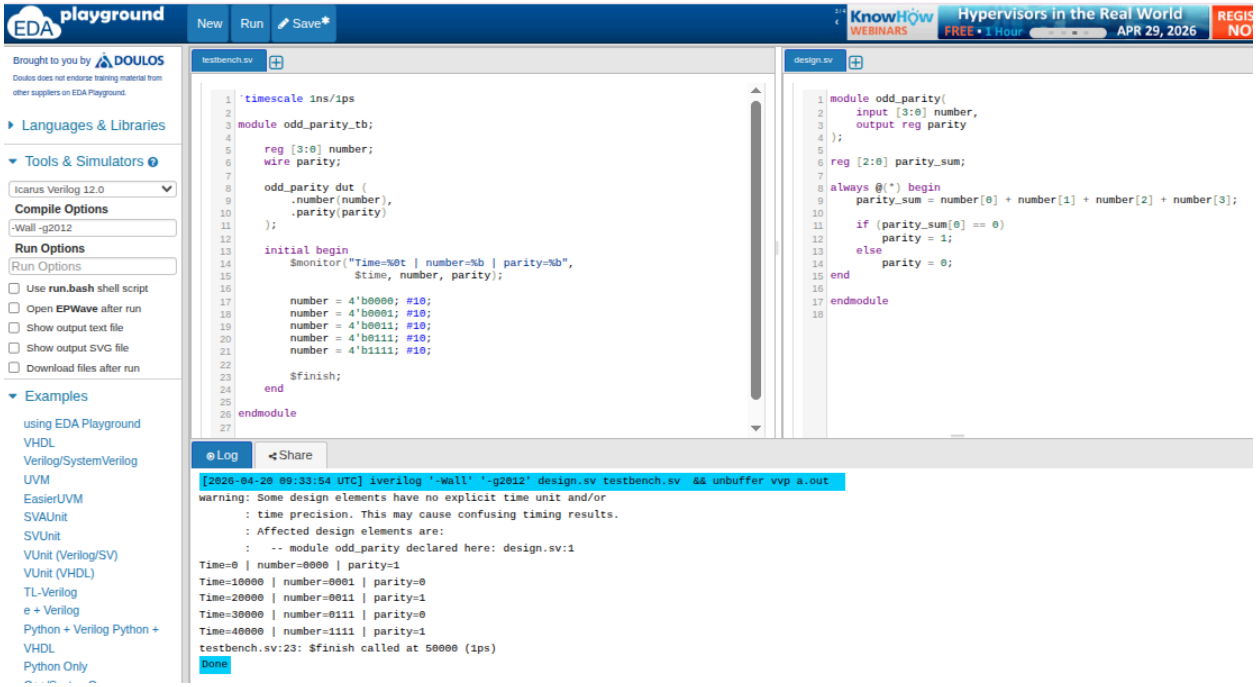
        number = 4'b0000; #10;
        number = 4'b0001; #10;
        number = 4'b0011; #10;
        number = 4'b0111; #10;
        number = 4'b1111; #10;

        $finish;
    end

endmodule
```


number
0:0 - 0:0
1:1 - 0:0
2:2 - 0:0
3:3 - 0:0
A
B
$2
$add
Y
$3
$add
$4
$add
parity_sum
1'1
1'0
0:0 - 0:0
S
$7
$mux
parity

odd_parity


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using EDA Playground
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Verilog/SystemVerilog
UVM
EasierUVM
SVAUnit
SVUnit
VUnit (Verilog/SV)
VUnit (VHDL)
TL-Verilog
e + Verilog
Python + Verilog Python + VHDL
Python Only
testbench.sv
design.sv
Log
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[2026-04-20 09:33:54 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
warning: Some design elements have no explicit time unit and/or
: time precision. This may cause confusing timing results.
: Affected design elements are:
: -- module odd_parity declared here: design.sv:1
Time=0 | number=0000 | parity=1
Time=10000 | number=0001 | parity=0
Time=20000 | number=0011 | parity=1
Time=30000 | number=0111 | parity=0
Time=40000 | number=1111 | parity=1
testbench.sv:23: $finish called at 50000 (1ps)
Done